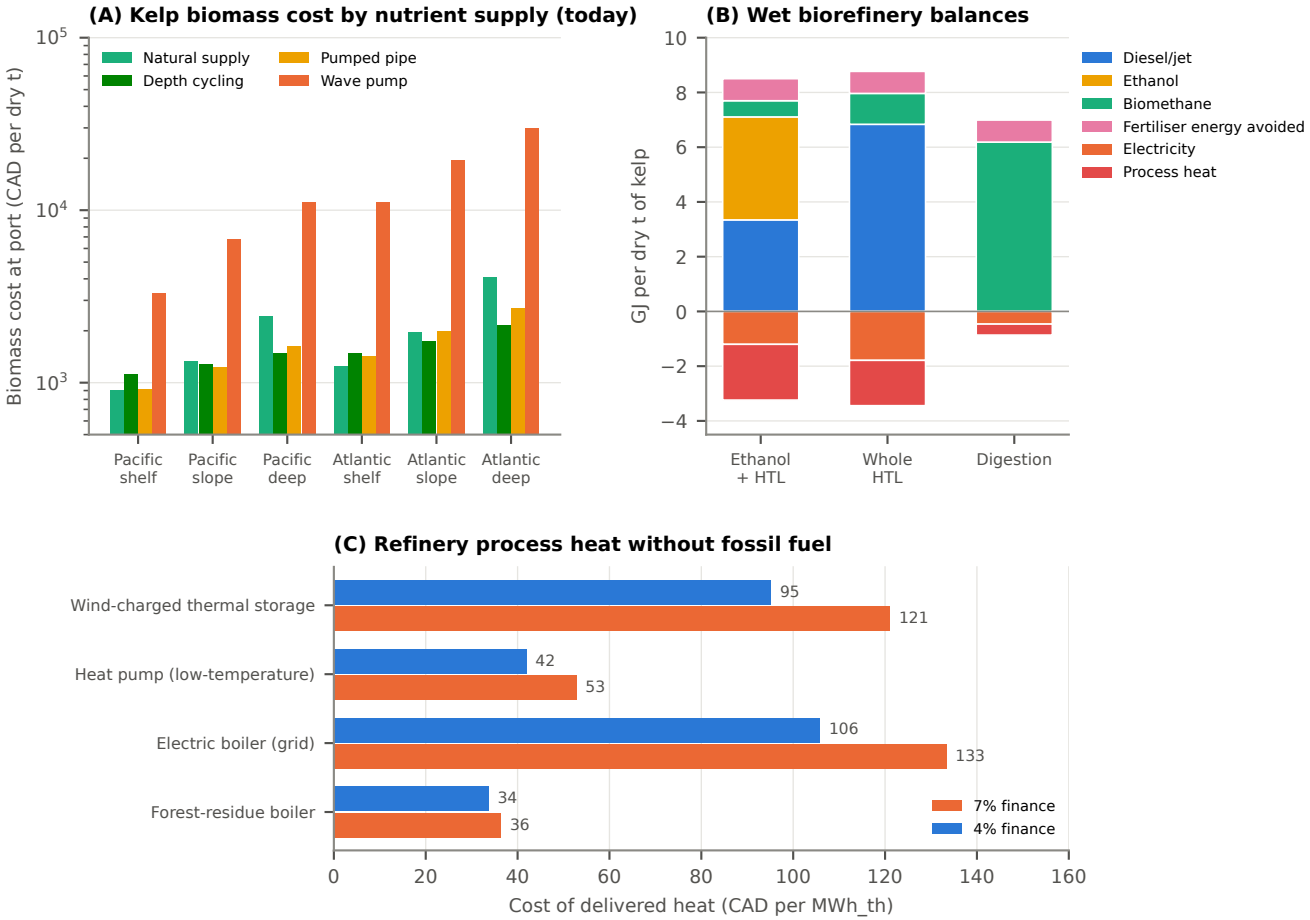
(A) Kelp biomass cost by nutrient supply (today)
Biomass cost at port (CAD per dry t)
10^5
10^4
10^3
Natural supply
Depth cycling
Pumped pipe
Wave pump
Pacific shelf
Pacific slope
Pacific deep
Atlantic shelf
Atlantic slope
Atlantic deep
(B) Wet biorefinery balances
GJ per dry t of kelp
10
8
6
4
2
0
−2
−4
Diesel/jet
Ethanol
Biomethane
Fertiliser energy avoided
Electricity
Process heat
Ethanol + HTL
Whole HTL
Digestion
(C) Refinery process heat without fossil fuel
Wind-charged thermal storage
95
121
Heat pump (low-temperature)
42
53
Electric boiler (grid)
106
133
Forest-residue boiler
34
36
7% finance
4% finance
0
20
40
60
80
100
120
140
160
Cost of delivered heat (CAD per MWh_th)